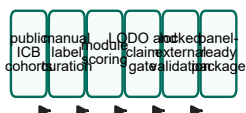


Fig. 1 | Multicohort ICB benchmark and leakage-safe study design

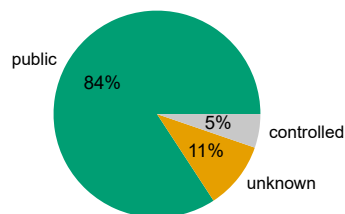
a

EcoNiche-Opt study workflow



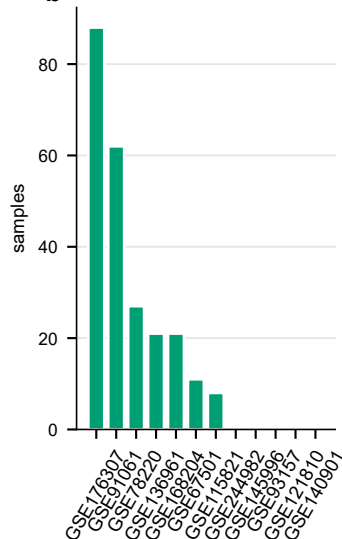
d

Data access status



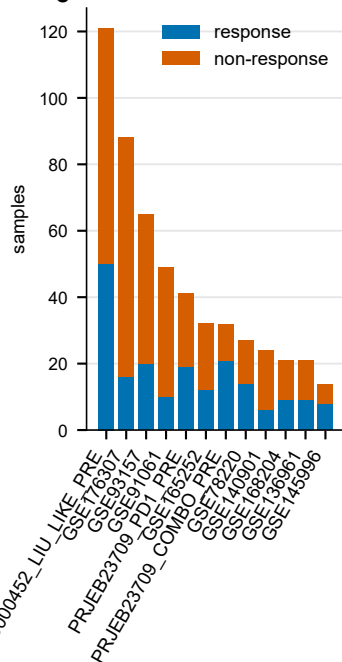
b

Processed cohorts



e

Response balance



C

Endpoint harmonization

strict REC	primary REC	clinical benefit
CR/PR=0	CR/PR/MR	PR/MR/SD=1
PD=0	SD/PD=0	PD=0
SD/MR=dro	binary	DCB/NDB

f

Module gene coverage

